Group Project

CRISPR Screening and Analysis

Online learning week

28 July - 1 August

Matt Coelho & Kalpana Surendranath 2025 WCS CRISPR

wellcome connecting science

CRISPR Screening and Analysis - Genome Informatics and Computational Design and Analysis

28 July - 1 August 2025, virtual via Zoom

wellcome sanger institute | UNIVERSITY OF WESTMINSTER

PRECOURSE ON LMS - Linux and R (for interest only)

Time	*Monday 28 July*	*Tuesday 29 July*	*Wednesday 30 July*	*Thursday 31 July*	*Friday 1 August*
08:45	Join Call	Join Call	Join Call	Join Call	Join Call
09:00	**Course introduction session** Wellcome Connecting Science, training team, participants	**Daily recap**	**Daily recap**	**Daily recap**	**Group project reminder** Matthew Coelho and Kalpana Surendranath
09:15					
09:30		**Genome engineering workflows and screening** Kalpana Surendranath	**Applications of base editing and prime editing in pooled and arrayed screens** Matthew Coelho	**ENCORE** Eloise Lockyer	**Group work time**
09:45					
10:00					
10:15					
10:30	**Break**	**Break**	**Break** (download data for next session)	**Break**	**Break**
10:45					
11:00	**Introduction to the CRISPR toolbox** Andrew Bassett	**Case studies** Munuse Savash Kalpana Surendranath	**Base editing guide RNA design** Cansu Dincer	**CRISPR screening informatics** Jamie Billington Saamin Cheema	**Group work time**
11:15					
11:30					
11:45	**Guided computational practical: genes and genomes** Saroor Patel				
12:00					
12:15					
12:30		**Q&A**	**Q&A**	**Q&A**	
12:45					
13:00	**Lunch**	**Lunch**	**Lunch**	**Lunch**	**Lunch**
13:15					
13:30					
13:45					
14:00	**Refresher quiz** Saroor Patel	**CRISPResso and validation of gene-edited models** Khalid Akram	**SNV and knock-in validation (Sanger sequencing - TIDE)** Sandeep Rajan	**Seminar** Manuel Kaulich	**Group presentations: Groups of 6 10 min presentation/ group and 5min QA**
14:15					
14:30	**Seminar** Alice Refermat				
14:45					
15:00				**Refresher quiz** Jamie and Saamin	
15:15					
15:30	**Intro to group project** Matthew Coelho, Kalpana Surendranath	**Refresher quiz** Kalpana and Khalid	**Refresher quiz** Cansu and Sandeep	**Project and practical Q&A**	**Wrap up and close**
15:45					
	Monday 28 July	*Tuesday 29 July*	*Wednesday 30 July*	*Thursday 31 July*	*Friday 1 August*

Virtual Groups

Group 1
Mehmet Bikkul
Ruhi Anjum
Zhe Yang
Marina Gysin
Katherine Diane Lugo Bardales

Group 2
Melanie Cook
Naima Azouzi
Claudia Maria Pinna
Kayani Kayani
Bo Fussing

Group 3
Andrew Curtis
Anam Razzak
Snir Boniel
Gratiela Gradisteanu
Salwa Lin
Ilaria Persico

Group 4
Bryony Kennedy
Zainab Bashir
Olayiwola Popoola
Marcos Magalhães
Mona Mahfood

Group 5
Iulia Maria Sabau
David Twesigomwe
Luigi Celauro
Shalini Kamu Reddy
Fatima Mohamedelmugadam

Group Project

A missense mutation in the coding region of the ***SRY* gene (g.2787144A>G, p.C154R)** has recently been found to alter the expression of the protein. This leads to an uncommon disorder in men.

The presence of the mutation does not always lead to disease, suggesting there are other factors at play. Your research team have been tasked with identifying genes involved in disease progression - particularly new drug targets to reverse the disorder.

Groups' tasks

1) Describe a CRISPR-based workflow to introduce the mutation in *SRY* in an appropriate cell model and verify the mutation.
2) Describe how you would design a CRISPR KO screen to identify modulators of *SRY* expression in this modified cell model.
3) Describe next steps after the screen, including a validation strategy.

*For each step, give a **rationale** for your approach and describe the **analysis** in each case.*

Modelling the *SRY* Promoter Mutation - Pointers

- What is your model system?
- Should it naturally express SRY? How about the efficiency of genetic manipulation?
- Your choice of CRISPR tools to introduce a precise genetic alteration
- Optimal method to provide correct copies of the DNA? How would you deliver the tools into the cells?
- What methods will allow you to screen for the correction?
- What methods can be used to measure both RNA and protein levels?

CRISPR Knockout Screen - Pointers

- How will you ensure the mutation is present? What factors will limit your analysis of more genes?
- What readout will allow you to separate cells with higher or lower expression?
- How will you sort and collect the most interesting/informative population of cells?
- What sequencing strategy will allow you to identify sgRNAs?
- How will you leverage the sequencing data to find candidate genes worth investigating further?

Group presentation

15 min per group

10 min for presentation and 5 min for questions

All the best!